

MODEL QUESTION PAPER

TED (10) 4029
[Revision – 2010]

Reg. No:.....
Signature:.....

SIXTH SEMESTER DIPLOMA EXAMINATION IN MECHANICAL ENGINEERING

REFRIGERATION AND AIR CONDITIONING

[Note : 1. Use of psychrometric chart allowed.
2. Missing data if any may be suitably assumed.]

[Maximum marks : 100]

[Time ; 3 Hours]

PART – A

[Maximum marks : 10]

Marks

I. Answer the following questions in one or two sentences:

1. Define the term 'Latent heat'
2. State the purpose of refrigeration.
3. What are the desirable properties of an ideal refrigerant ?
4. Show the dehumidification process on a psychrometric chart.
5. What do you understand by 'Cryogenics'

[5 x 2 = 10]

PART – B

[Maximum marks : 30]

II. Answer any **five** questions:

1. List the various fields of applications of refrigeration.
2. What is the effectiveness of a refrigeration system ? How is it measured ?
3. Explain the working of dry expansion evaporator.
4. Derive the efficiency of heating and cooling coils.
5. Explain the construction and using of psychrometric charts.
6. List the factors affecting optimum effective temperature.
7. Explain Cascade refrigeration.

[5 x 6 = 30]

PART – C

[Maximum marks : 60]

[Answer **one full question** from each module]

MODULE – I

- III. a). Draw the p – V and T – S diagrams of a reversed Carnot cycle applied to a refrigerating machine. [4]
- b). Explain the effect of sub-cooling and super-heating the refrigerant. [6]
- c). In an ammonia vapour compression system the pressure in the evaporator is 2 bar. Ammonia at exit is 0.85 dry and at entry it is 0.19 dry. During the compression the work done per kg of ammonia is 150 kJ. Calculate the COP. The latent heat at 2 bar is 1325 kJ/kg. [5]

OR

- IV. a). Explain the merits and demerits of air refrigeration system. [5]
b). A refrigerator working on Bell – Coleman cycle operates between pressure limits of 1.05 bar and 8.5 bar. Air is drawn from the cold chamber at 10^0 compressed and then it is cooled to 30^0C before entering the expansion cylinder. The expansion and compression follows the law $pv^{1.3} = \text{constant}$. Determine the theoretical COP of the system. [10]

MODULE – II

- V. a). List the various fields of applications of refrigeration. [4]
b). Draw the neat sketch of domestic refrigerator. [6]
c). Explain with suitable sketch, the working of thermostatic expansion valve. [5]

OR

- VI. a). Compare the vapour absorption system with vapour compression system. [8]
b). Explain the ice cream manufacturing process. [7]

MODULE – III

- VII. a). Explain and represent the process on psychrometric chart, heating and dehumidifying [5]
b). Find the properties of humid air whose dry bulb temperature is 33^0C and wet bulb temperature is 21^0C . [10]

OR

- VIII. a). Sketch the basic psychrometric process on a single psychrometric chart. [5]
b). 900 kg/hour of return air at a DBT 24^0C and RH 60% mixes with 100 kg/hour of fresh air of DBT 40^0C and RH 30%. Calculate the final condition of the mixture. [10]

MODULE – IV

- IX. a). Write short notes on occupants heat gain. [4]
b). Explain with neat sketch the working of winter air conditioning system. [6]
c). Sketch a neat layout of window air conditioner. [5]

OR

- X. a). Write short notes on product load. [5]
b). A hotel dining hall is required to be air conditioned. Following data is available

Dimensions of the dining hall = 15 X 15 X 10 m

Heat flows through walls, roof and floor = 12600 kJ/hour

Solar heat gain through glass = 6300 kJ/hour

Equipment sensible heat gain = 4200 kJ/hour

Equipment latent heat gain = 2100 kJ/hour

Quantity of fresh air = $2500 \text{ m}^3/\text{hour}$

Quantity of infiltration air = $500 \text{ m}^3/\text{hour}$

Fresh air load:

a) Sensible – 10500 kJ/hour

b) Latent – 12600 kJ/hour

Infiltration air load:

a) Sensible –2100 kJ/hour

b) Latent –4200 kJ/hour

Heat load per person:

a) Sensible –210 kJ/hour/occupant

b) Latent –315 kJ/hour/occupant

The dining hall has the capacity to accommodate 50 persons.

The conditions of air may be assumed as;

Outside – 40⁰C DBT, 75% RH

Designed inside – 25⁰C DBT, 50% RH

Determine

a). Room sensible heat factor, and

b).Grand sensible heat factor.

[10]
